A conditional obstruction to polynomial sample complexity in unrestricted NNbayes

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Abstract

We study exact Bayesian prediction with fully connected tanh networks, a linear readout, and independent centered Gaussian weights. For input dimension d, depth L, and hidden width N, fix a prescribed scale schedule $t = t(d, L, N) \in (0, 1]$; at fixed d, L, this is a function $t(N)$ of learner width. The weight variances are t^2/d in the first layer, t^2/N in intermediate layers, and t^2/N^2 in the readout. The posterior conditions on an absolute output margin greater than $1/2$ at every Boolean input and exact agreement with noiseless training labels. A test input counts as correctly predicted only when its correct label has posterior probability greater than $2/3$.

Assume that polynomial-size Boolean circuits cannot all be replaced by polynomial-size, logarithmic-depth threshold circuits: $\mathsf{P/poly} \neq \mathsf{TC}^1$, with both classes nonuniform. Then this learner has no distribution-free sample bound jointly polynomial in input dimension d, teacher complexity C, and depth L that holds at every learner width $N \geq C$. Here C bounds both the teacher's hidden width and all its weight magnitudes, and L is the common depth of teacher and learner. The conclusion holds for every prescribed positive scale schedule, including every power-law scale $t = N^{(1-\gamma)/2}$ with $\gamma \geq 1$, already at fixed accuracy and confidence. Inference may use unlimited computation.

The proof shows that, at sufficiently large finite width and sufficient depth, most posterior mass lies on functions computable by small, logarithmic-depth threshold circuits. A finite majority argument turns distribution-free learning by mixtures of these functions into an exact threshold-circuit representation of every polynomial-size Boolean computation, contradicting the complexity assumption. The main theorem, including the required finite-width posterior bound, is proved in Lean. The conclusion rules out a universal polynomial guarantee; it does not establish an exponential lower bound in depth.

1 Introduction

A natural goal for Bayesian neural-network learning (NNbayes) is a sample bound polynomial in the size of a circuit implementing the target, independently of the much larger network used by the learner. At each fixed depth, Gaussian tanh priors at mean-field scaling and stronger scales admit such width-independent bounds. The degree or constants of those bounds can grow rapidly with depth. It is therefore important to distinguish a polynomial bound *at every fixed depth* from one polynomial jointly in width and depth.

This paper gives a conditional negative answer to the latter question for the Gaussian prior defined in Section 2, without an additional robustness restriction. Under the circuit-complexity conjecture $\mathsf{P/poly} \neq \mathsf{TC}^1$, a universal joint-polynomial sample bound is impossible. This is a

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statistical obstruction, even for exact Bayesian inference with unlimited computation. It does not arise from the difficulty of running a posterior-sampling algorithm.

The mechanism is a mismatch between target expressivity and posterior expressivity at limited sample counts. Every small Boolean circuit can be implemented by a polynomial-resource tanh teacher with bounded weights. An analytic theorem for this posterior nevertheless places most posterior mass on functions having threshold circuits of size polynomial in dimension, sample count, and learner depth, and of logarithmic threshold depth. If the learner succeeded for every input distribution using only polynomially many examples, an elementary boosting argument would turn these posterior mixtures into an exact TC^1 circuit for every function in P/poly.

The width quantifier is essential: the guarantee must cover arbitrarily large finite learner widths. So is the distribution quantifier: training and testing use the same distribution, but the guarantee must hold for every such distribution. No distribution shift is involved.

Section 2 defines the learner, teacher resources, and circuit classes. We then state the conditional impossibility theorem and prove it by combining a posterior circuit bound with finite boosting and a bounded-weight implementation of Boolean circuits. The analytic bound is the part of the argument requiring substantial background: Section 4 states it with its full quantifiers, and Appendix A explains its proof. The detailed entropy and approximation estimates are developed in [1]; the accompanying Lean proof includes the instances needed here. The statistical and circuit-complexity reduction is given in full below.

2 Learner, target complexity, and circuit classes

2.1 The Gaussian tanh posterior

Let $d \geq 1$, $X_d = \{-1,1\}^d$, and $f : X_d \to \{-1,1\}$. A network has weighted depth $L \geq 3$, hidden width N, augmented input $h^0(x) = (1,x)$, hidden states

$$h^j(x) = \tanh(W^j h^{j-1}(x)), \qquad 1 \leq j < L,$$

and scalar linear output $z_W(x) = W^L h^{L-1}(x)$. The activation acts coordinatewise. There are no separate hidden biases; constant wires use ordinary neurons and are counted in the width.

For $0 < t \leq 1$, let $\Pi_{d,L,N,t}$ be the product Gaussian law on all weights, with mean zero and entrywise variances

$$\frac{t^2}{d} \quad \text{in the first layer,} \qquad \frac{t^2}{N} \quad \text{in intermediate layers,} \qquad \frac{t^2}{N^2} \quad \text{in the readout.} \tag{1}$$

The critical, or mean-field, scale is $t = 1$. The usual exponent family $\gamma \geq 1$ is obtained by setting $t = N^{(1-\gamma)/2}$.

Fix a prescribed schedule

$$t = t(d, L, N) \in (0, 1]. \tag{2}$$

It may depend on these architectural dimensions but not on the training sample or the input distribution. It need not be computable.

For a labeled sample $S = ((x_i, f(x_i)))_{i=1}^n$, condition the prior on

$$|z_W(x)| > 1/2 \quad (x \in X_d), \qquad f(x_i) z_W(x_i) > 1/2 \quad (1 \leq i \leq n). \tag{3}$$

Let Π_S denote this posterior and put

$$p_S(y \mid x) = \Pi_S\{W : \operatorname{sign} z_W(x) = y\}.$$

For an input law μ on X_d, the confidence-failure mass is

$$\mathcal{E}_\mu(f, S) = \mu\{x : p_S(f(x) \mid x) \le 2/3\}. \tag{4}$$

Thus a prediction counts as successful only when the posterior assigns probability strictly greater than $2/3$ to the correct label. The training inputs are iid from μ, which is also the test law.

Here "unrestricted" means that no internal-stability or other robustness condition is added to the prior support. The global absolute-margin condition in (3) remains part of the learner.

2.2 Teacher resources and universal polynomial learning

For an integer $C \ge 1$, a target has teacher resources (d, C, L) if it admits a depth-L tanh network of hidden widths at most C, all weights bounded in absolute value by C, and signed output greater than $1/2$ on every input. Equivalently, its hidden widths may all be taken equal to C by zero padding. Larger hidden widths can also be obtained by zero padding. Continuity on the finite cube then gives a positive-prior neighborhood of that implementation, so the conditional law (3) is defined for every sample and every $N \ge C$.

For fixed statistical tolerances, any polynomial bound in d, C, L is bounded above by $A(1 + d + C + L)^k$ for some real $A > 0$ and integer $k \ge 0$. We use that envelope to state the question precisely.

Definition 2.1 (Universal polynomial sample guarantee)**.** *For a schedule (2), let* $\mathsf{PolyLearn}(t)$ *mean that there are $A > 0$ and $k \in \mathbb{N}$ such that, for every $d \ge 1$, $C \ge 1$, $L \ge 3$, every target with those teacher resources, every $N \ge C$, every law μ, and every integer sample count*

$$n \ge A(1 + d + C + L)^k, \tag{5}$$

an iid labeled sample satisfies

$$\mathbb{P}_{S\sim\mu^n}\{\mathcal{E}_\mu(f, S) \le 1/16\} \ge 1/2. \tag{6}$$

The constants A, k are independent of learner width and input law.

A polynomial guarantee at all accuracy and confidence parameters would in particular imply (6). The proof below uses the particular integer count $n = A_0(1 + d + C + L)^k$, with a fixed positive integer $A_0 \ge A$; it does not assume that posterior accuracy is monotone in n.

The quantity $d + C + L$ is polynomially equivalent to an ordinary upper bound on the number of parameters in this dense teacher architecture, after including input dimension and depth. More importantly, the reduction below starts with ordinary Boolean circuits and uses constant-magnitude integer weights. The obstruction does not require hiding an arbitrary truth table in the precision of a teacher coefficient.

2.3 Nonuniform threshold circuits

A circuit family has one Boolean circuit for each input dimension. "Nonuniform" imposes no efficient procedure for generating that family. The class $\mathsf{P/poly}$ consists of functions with polynomial-size Boolean circuit families. The class TC^1 consists of functions with polynomial-size threshold circuit families of depth $O(\log(d+2))$. One may use unbounded-fan-in majority gates and ordinary Boolean gates as the threshold basis. The class TC^0 uses constant depth instead. Threshold-circuit size counts gates plus their incoming wires. All circuit classes in this paper are nonuniform.

The assumption $\mathsf{P/poly} \ne \mathsf{TC}^1$ says that some polynomial-size computations cannot be parallelized into logarithmic threshold depth with polynomial size. Appending a constant number of polynomial-fan-in majority layers preserves TC^1. The reverse simulation has no depth requirement:

Lemma 2.2 (Threshold circuits are polynomial-size Boolean computations)**.** *A threshold circuit with total gate and input-wire count s has an ordinary Boolean circuit of size polynomial in s. Consequently* $\mathsf{TC}^1 \subseteq \mathsf{P/poly}$.

Proof. For a gate with m incoming bits, define $T_{j,k}$ to mean that at least k of the first j bits are one. The recurrence

$$T_{j,k} = T_{j-1,k} \ \mathsf{OR}\ (x_j \ \mathsf{AND}\ T_{j-1,k-1})$$

with $T_{j,0} = 1$ and $T_{0,k} = 0$ for $k > 0$ computes every relevant count threshold using $O(m^2)$ gates. Thresholds above m are constant false. Replacing every threshold gate this way costs at most a constant times $\sum_j (m_j + 1)^2$, which is $O((s+1)^2)$. AND, OR, and NOT have constant-size NAND implementations. No bound on the Boolean simulation's depth is needed for membership in $\mathsf{P/poly}$. □

3 The conditional impossibility theorem

Theorem 3.1 (No universal polynomial circuit-learning bound)**.** *Assume* $\mathsf{P/poly} \neq \mathsf{TC}^1$. *For every prescribed scale schedule $t(d, L, N) \in (0, 1]$, the property* $\mathsf{PolyLearn}(t)$ *is false.*

Equivalently, for every real $A > 0$ and integer $k \geq 0$, there exist dimensions d, C, L, a target with those teacher resources, a learner width $N \geq C$, a common training/test law μ, and an integer sample count satisfying (5), such that

$$\mathbb{P}_{S\sim\mu^n}\{\mathcal{E}_\mu(f, S) > 1/16\} > 1/2. \tag{7}$$

In particular the conclusion holds separately for every fixed $\gamma \geq 1$ in the exponent family.

Theorem 3.1 concerns unlimited-computation Bayesian prediction. Its target family may be chosen from $\mathsf{P/poly}$, with polynomial-resource bounded-weight teachers. The difficult distribution and the learner width may depend on the proposed sample polynomial.

Corollary 3.2 (What must change)**.** *Under the same complexity assumption, no theorem can give the learner of Section 2 a sample guarantee polynomial jointly in d, C, L at every eligible learner width, for every target in Section 2 and every common training/test law. Consequently, obtaining such a guarantee while retaining the architecture, that target class, those statistical conventions, and those width and distribution quantifiers requires changing the prior. Restricting its support to networks satisfying an additional stability condition is one possible change, not a conclusion that every such restriction suffices.*

If other parts of the goal may change, there are other routes: restrict the target class, assume a particular input law, limit learner width, or alter the likelihood or prediction criterion. The theorem does not say that a robustness restriction is the uniquely necessary remedy.

4 The analytic input: a posterior circuit ceiling

For integers $R \geq 2$, let $\mathcal{H}_R$ be the finite set of Boolean functions representable by threshold circuits of size at most R^b and depth at most $b \log R$, for a sufficiently large universal constant b. The particular circuit for a function need not be unique.

Proposition 4.1 (Posterior circuit ceiling)**.** *There are universal positive constants b, c_0, c_1 such that the following holds for every integer sample budget $n \geq 1$ for the posterior (1)–(3). Put $R = d + n + L + 2$ and suppose*

$$L \geq c_0 \log R. \tag{8}$$

For every $0 < \beta < 1$, every actual integer width

$$N \geq 2^{c_1 d} R^{c_1} + c_1 (L-1) \log(1/\beta), \tag{9}$$

every $0 < t \leq 1$, and every consistent sample of at most n observations whose conditioning event has positive probability,

$$\Pi_S\{W : \operatorname{sign} z_W \in \mathcal{H}_R\} \geq 1 - \beta. \tag{10}$$

The class $\mathcal{H}_R$ is common to all these samples, widths, and scales. In particular, the width threshold is independent of t and of the input law.

Proposition 4.1 is the uniform finite-width posterior circuit bound of [1]. Its proof has three ingredients. First, finite-event relative-entropy certificates associate most posterior mass with controlled covariance paths of bounded action. A target-mass construction for a sample memorizer supplies the posterior denominator needed for this step. Second, the bounded-action paths admit finite canonical feature representations, with bounded-size covers even when t approaches one. Third, parallel iteration of the resulting feature recursion gives polynomial circuit size and logarithmic threshold depth. The actual finite-width estimate, rather than only a limiting variational statement, is essential to (10).

The present proof uses Proposition 4.1 only with $\beta = 1/6$. It does not use the explicit covariance-path rate, an optimal representation, or an efficient way of constructing a circuit from network weights. Nor does it assert that one circuit carries most posterior mass. The posterior can be an arbitrary mixture over its ceiling class.

5 From distribution-free mixtures to an exact circuit

The next lemma is independent of neural networks. It is an elementary finite boosting argument. Its deterministic weight update avoids any need for a minimax theorem or a concentration inequality.

Lemma 5.1 (A small exact majority from weak prediction)**.** *Let X be a finite nonempty set with $|X| \leq 2^d$, let $f : X \to \{-1, 1\}$, and let $\mathcal{H}$ be a nonempty finite class of Boolean functions. Suppose that for every probability law μ on X there is an $h \in \mathcal{H}$ with*

$$\mu\{x : h(x) \neq f(x)\} \leq 9/20.$$

Then f is exactly a majority of $O(d+1)$ members of $\mathcal{H}$, allowing repetitions. In fact $400(d+1)$ voters suffice.

Proof. Initially give every input weight $w_0(x) = 1$. At step j, normalize the positive weights to a probability law and choose a voter h_j whose weighted error is at most $9/20$. Multiply a correctly classified input's weight by $9/10$, and an incorrectly classified input's weight by $10/9$. If $W_j = \sum_x w_j(x)$ is the total weight, then

$$\frac{W_{j+1}}{W_j} \leq \frac{11}{20}\frac{9}{10} + \frac{9}{20}\frac{10}{9} = \frac{199}{200}.$$

After m rounds,

$$W_m \le |X|(199/200)^m \le 2^d e^{-m/200}. \tag{11}$$

Set $m = 400(d+1)$. For an elementary rational check, Bernoulli's inequality gives $(200/199)^{400} \ge 1 + 400/199 > 2$, so $(199/200)^{400} < 1/2$. Consequently $W_m \le 1/2 < 1$. But an input misclassified by at least half the voters would have final weight

$$(9/10)^{\#\text{correct}}(10/9)^{\#\text{wrong}} \ge 1,$$

contradicting $W_m < 1$. Every input is therefore correctly classified by a strict majority of the chosen voters. □

Lemma 5.2 (A successful posterior supplies a weak voter)**.** *Let π be a distribution on Boolean functions on X, and suppose $\pi(\mathcal{H}) \ge 5/6$. If*

$$\mu\{x : \pi\{h : h(x) = f(x)\} \le 2/3\} \le 1/16,$$

then some $h \in \mathcal{H}$ has μ-error at most $9/20$.

Proof. At every point outside the confidence-failure set, a draw from π has wrong-label probability less than $1/3$. On that set its error probability is at most one. Averaging over inputs gives

$$\mathbb{E}_{h\sim\pi}\mu\{x : h(x) \ne f(x)\} \le \frac{1}{3} + \frac{2}{3}\frac{1}{16} = \frac{3}{8}.$$

Conditioning on $\mathcal{H}$ increases this average by at most $6/5$, so its conditional average is at most $(3/8)(6/5) = 9/20$. A finite average cannot be smaller than all its contributing terms. One $h \in \mathcal{H}$ therefore has error at most $9/20$. □

Proposition 5.3 (Posterior success forces a small parallel circuit)**.** *Fix d, n, L satisfying (8), one width satisfying (9) with $\beta = 1/6$, and one scale $0 < t \le 1$. These choices are made before the input distribution. Suppose that for every input law μ, there is positive probability of an iid labeled n-sample with $\mathcal{E}_\mu(f, S) \le 1/16$. Then f has a threshold circuit of size polynomial in $d + n + L$ and depth $O(\log(d+n+L+2))$. The constants are universal and independent of learner width and scale.*

Proof. For each law μ, choose a successful sample and push its posterior forward to Boolean functions. Proposition 4.1 gives mass at least $5/6$ to the common finite class $\mathcal{H}_R$. Lemma 5.2 supplies one member of that class with μ-error at most $9/20$.

This is precisely the distribution-free hypothesis of Lemma 5.1. Its exact majority has $O(d+1)$ voters from $\mathcal{H}_R$. Compile the voters in parallel and append one majority gate. Size grows by only a polynomial factor, and depth increases by one. □

The posterior mixture is handled by averaging; it is not identified with one selected predictor. The successful samples and chosen voters may vary with the input law, while $\mathcal{H}_R$ remains fixed. Positive sample-success probability is used only to supply one successful sample for each distribution. An earlier two-majority/minimax proof in [1] permits every confidence-failure tolerance below $1/2$; the smaller fixed tolerance here makes the proof substantially shorter.

6 Small Boolean circuits have bounded-weight teachers

Lemma 6.1 (Digital tanh implementation)**.** *Every NAND circuit with s gates on d inputs has a tanh implementation of hidden width $C = d + s + 256$ and weighted depth $L = d + s + 3$, with all weights integers of absolute value at most 192 and signed output greater than $1/2$ throughout X_d. It has no hidden biases. The depth can be further increased by inserting copy layers with the same bounds. Other fixed bounded-fan-in Boolean bases incur only a constant-factor change in circuit size before this construction.*

Proof. Use signed Boolean wire values. NAND, NOT, and COPY have the scores

$$1 - u - v, \qquad -u, \qquad u.$$

Their signed margins are at least one on exact Boolean inputs. Reserve coordinates for the d inputs, the two Boolean constants, and every computed gate. Initialize their tanh encodings in one hidden layer from the augmented input. Then process one NAND gate per layer, retaining all earlier values by copying. The true constant wire supplies all subsequent affine constants. Append $d + 1$ copy layers. The number of hidden-to-hidden stages is $s + d + 1$, hence the weighted depth is $s + d + 3$. The width $C = d + s + 256$ leaves all required coordinates available.

Use gain 64 at each gate. Suppose that each incoming wire is within $1/16$ of its intended sign. The score perturbation, including the constant wire, is at most $3/16$. Thus the signed preactivation is at least $64/2$. The elementary saturation bound

$$by \geq A/2, \quad b \in \{-1, 1\}, \quad A > 0 \quad \Longrightarrow \quad |\tanh y - b| \leq 4/A$$

then gives a new error at most $1/16$. For example, the bound follows from $1 - \tanh v = 2/(e^{2v} + 1)$ and $e^a \geq 1 + a$. The initial input and constant signs satisfy the same estimate. This proves an error invariant uniform in the number of layers.

A coefficient receives at most three contributions of magnitude 64, including coincident input and constant wires, so its absolute value is at most 192. The final linear readout is twice the selected output wire. Its signed margin is at least $2(1 - 1/16) = 15/8 > 1$. Further copy layers preserve the invariant. Unused neurons have zero weights, so zero padding to a larger width also preserves the function. □

The teacher construction is digitally stable at its Boolean gate margins. The conditional negative result nevertheless concerns the *unrestricted learner prior*; having a stable implementation of the target does not force the posterior's competing explanations to share that stability.

7 Proof of the main theorem

We prove the contrapositive in its slightly stronger positive form.

Theorem 7.1 (A universal polynomial guarantee would collapse classes)**.** *For every schedule* (2)*,*

$$\mathsf{PolyLearn}(t) \quad \Longrightarrow \quad \mathsf{P/poly} = \mathsf{TC}^1.$$

Proof. Assume Definition 2.1, with fixed A, k, and choose a positive integer $A_0 \geq A$. Take any family (f_d) in $\mathsf{P/poly}$, represented by Boolean circuits of size polynomial in d. Lemma 6.1 gives teachers with width, weight bound, and depth polynomial in d. Their width C_d is at least 256, so it also bounds every coefficient, and their depth L_d is already at least d.

The prescribed sample count

$$n_d = A_0(1 + d + C_d + L_d)^k$$

is polynomial in d and is fixed *before* choosing learner width. Thus $R_d = d + n_d + L_d + 2$ is polynomial in d. Since $L_d \geq d$, the padding condition $L_d \geq c_0 \log R_d$ holds for all sufficiently large d.

For each such dimension choose one integer width at least C_d and at least the threshold (9) with $\beta = 1/6$. The threshold is independent of t, so now set $t_d = t(d, L_d, N_d)$. There is no circular choice of width and scale. Both are fixed before the input law.

The assumed learning guarantee gives success probability at least $1/2$ for every input law. Proposition 5.3, applied with confidence-failure tolerance $1/16$, produces a threshold circuit for f_d of polynomial size in R_d, hence polynomial size in d, and depth $O(\log R_d) = O(\log(d + 2))$. The constants are uniform in d. Finitely many remaining input dimensions can be hardcoded.

Every family in P/poly therefore belongs to nonuniform TC^1. The reverse inclusion is Lemma 2.2, so $\mathsf{P}/\mathrm{poly} = \mathsf{TC}^1$. □

Taking the contrapositive proves Theorem 3.1. For fixed A, k, a violation of (6) has probability strictly less than $1/2$ for the event $\mathcal{E}_\mu(f, S) \leq 1/16$. Its complementary event is exactly (7). All distributions and samples here are on a finite input set, so no measurability issue is hidden in this logical step.

The reduction has the following order of choices:

$$\begin{aligned} &\text{sample polynomial} \longrightarrow \text{target family and teachers} \longrightarrow n_d, \\ &n_d \longrightarrow N_d \longrightarrow t_d \longrightarrow \text{input law and sample.} \end{aligned}$$

The last two objects may vary while the architecture and scale remain fixed. A sample bound polynomial in N would not permit this argument: it would make n_d depend on the large width chosen only afterwards.

7.1 A fixed hard family and one law for all sufficiently large widths

The argument gives more than the failure of a particular confidence threshold. The following consequence separates what can be made uniform from the absence of an explicit asymptotic lower-bound rate.

Corollary 7.2 (Failure for every sample on a hard distribution)**.** *Assume* $\mathsf{P}/\mathrm{poly} \neq \mathsf{TC}^1$. *There is a fixed family* (f_d) *in* P/poly *with bounded-weight tanh teachers of width* $C_d = \mathrm{poly}(d)$ *and depth* $d \leq L_d = \mathrm{poly}(d)$ *such that the following holds. For every polynomially bounded integer sample budget* $n_d \geq 1$*, there are infinitely many dimensions* d *with an input law* μ_d *for which*

$$\mathcal{E}_{\mu_d}(f_d, S) > 1/16$$

for every labeled sample S *of length at most* n_d*, at every width above both* C_d *and the analytic ceiling threshold, and every scale* $0 < t \leq 1$*. The law* μ_d *is common to all those samples, widths, and scales. In particular, the probability of failure is one for iid sampling from* μ_d *at any of those sample counts.*

Proof. Fix any family in $\mathsf{P}/\mathrm{poly} \setminus \mathsf{TC}^1$ and compile its teachers as in the proof of Theorem 7.1. Put $R_d = d + n_d + L_d + 2$. For all sufficiently large d, the padding condition holds. Let $\mathcal{K}_d$ be the class of all majorities of $400(d + 1)$ members of $\mathcal{H}_{R_d}$, allowing repetitions. Every member has size polynomial in R_d and depth $O(\log R_d)$. Since $R_d = \mathrm{poly}(d)$, if $f_d \in \mathcal{K}_d$ for all but finitely many d, the entire family would be in TC^1. Hence $f_d \notin \mathcal{K}_d$ for infinitely many d.

At each such dimension, the contrapositive of Lemma 5.1 gives one law μ_d for which every member of $\mathcal{H}_{R_d}$ has error strictly greater than 9/20. Proposition 4.1 applies to every eligible width, scale, and consistent sample of at most n_d points, using the same class $\mathcal{H}_{R_d}$. By Lemma 5.2, none of those posteriors can have confidence-failure mass at most 1/16 under μ_d. This proves the claim. All target-labeled samples have positive conditioning probability because the teacher exists. □

The teacher family is fixed before the sample budget. The hard input law may depend on the budget and dimension. No particular growth rate or explicit description of the hard family follows from the separation assumption alone.

8 Interpretation and limits

What "no polynomial bound" means. The conclusion defeats every universal polynomial resource bound. It does not assert an exponential lower bound in L, or even a fixed stretched-exponential rate. Quasipolynomial or other intermediate sample growth remains compatible with the theorem. The input dimension and teacher resources vary together; fixing the finite input domain would give an elementary finite-domain upper bound regardless of depth.

Why the input distribution is allowed to vary. For each proposed polynomial, the hard instance may use its own common training/test law. The boosting argument would not follow from success only under the uniform distribution. The theorem therefore supplies no uniform-input lower bound without additional work.

No contradiction with positive fixed-depth theorems. For each fixed L, a bound with degree depending on L can hold. The theorem prohibits one degree working jointly over all depths and target resources. It also leaves intact width-dependent finite-width bounds and positive bounds for particular target representations.

Robust prior restrictions. Positive joint-polynomial results obtained by discarding unstable networks change the learner. Theorem 3.1 gives a conditional reason that some change is needed for a theorem with the target class and width and distribution quantifiers of Definition 2.1. It does not identify the best restriction, establish that trained networks satisfy one, or prove that support restriction is preferable to every other modification.

Computational and statistical claims. The use of a circuit-complexity conjecture does not make this a lower bound on the runtime of inference. The posterior is the exact conditional Gaussian law. Circuit complexity enters because of the functions to which that posterior assigns mass, and because distribution-free prediction can be converted into an exact finite circuit.

Formal verification scope. The accompanying Lean development proves Theorem 3.1 for the actual Gaussian prior, tanh network, global-margin posterior, iid sample law, and concrete Boolean and threshold circuits defined above. Its final proved proposition, `ConditionalImpossibility`, states $\mathsf{P}/\mathrm{poly} \neq \mathsf{TC}^1 \Rightarrow \forall t\, \neg\mathsf{PolyLearn}(t)$ with exactly these conventions. The equivalent finite counterexample formulation is also proved. The complexity separation remains an assumption; the posterior circuit ceiling is a proved dependency, not an additional hypothesis of the final theorem. The scale quantifier covers positive finite scales; it does not assert the separately defined zero-temperature posterior obtained by taking a limit at fixed width.

The checked proof includes the finite-width target-mass construction, conditional-entropy path extraction, all three scale regimes described in Appendix A, finite arithmetic and threshold-circuit compilation, bounded-weight teacher compilation, posterior-to-weak-voter transfer, finite boosting, and the circuit-family implication. Its axiom reports contain only Lean's standard `propext`, `Classical.choice`, and `Quot.sound`.

This verification claim concerns the main theorem. The analytic ceiling is formalized for the instances needed in the reduction: $\beta = 1/6$, integer resource envelopes, a target with an eligible realizing teacher, and samples of the specified length. The stronger statement of Proposition 4.1 for every $\beta \in (0, 1)$ and the uniform hard-family conclusion of Corollary 7.2 are not claimed to be formalized here. The verification record identifies the final entry point, exact checked sources, and reproducible build.

A How the analytic posterior ceiling is obtained

This appendix describes the analytic argument needed for the main theorem, at the fixed exceptional mass $\beta = 1/6$. The scalar entropy and approximation estimates used below are proved in [1]; the accompanying Lean development also proves their required instances and constructs the final finite threshold circuits.

From posterior mass to a controlled path. Write $H = L - 1$, $\ell = \log R$, and

$$a = t^{-2} - 1, \qquad c_t = \frac{a - \log(1 + a)}{2}.$$

A controlled feature path records the Gram matrices of all Boolean inputs through the hidden layers. Its layer law has relative entropy D_j with respect to the corresponding Gaussian row law. If b is its final linear readout, its action is

$$\mathcal{A} = \sum_{j=1}^{H} D_j + \frac{\|b\|^2}{2t^2}.$$

The path realizes the same Boolean function with strict signed margin $1/2$. A sample-memorizer construction lower-bounds the actual finite-width conditioning probability by $\exp(-NA_0)$, where

$$A_0 \le Hc_t + V_* + B\left(a\ell + \frac{\ell^2}{H}\right)$$

for a universal B. Here V_* is a universal finite constant: the baseline cost of the one-dimensional variance-control problem in [1]. It is independent of dimension, sample count, depth, and scale. This construction applies to every target-labeled sample; it does not require the target itself to have a small representation.

Conditioning the weight law on a sufficiently massive event gives a finite relative-entropy budget. Conditional entropy decomposition, finite-dimensional approximation, and margin repair then extract a genuine controlled path from that event. Choose extraction slack of order $1/(t^2H)$. The repair width is polynomial in the *normalized* action t^2A_0 and the other resources, with the factor exponential in d already allowed in (9). Thus its width threshold is uniform in t. If more than $1/6$ of posterior mass lay outside the proposed circuit class, this extraction would produce an outside-class path with

$$\mathcal{A} \le Hc_t + V_* + B'\left(a\ell + \frac{\ell^2}{H}\right). \tag{A.1}$$

The remaining task is to compile every such path into the common class.

Small raw action, including the critical endpoint. A scalar occupation estimate controls paths satisfying $\mathcal{A} \le V_* + O(\ell^2/H)$. Adaptive Gaussian feature replacement reduces their representations to polynomially many coordinates. After normalization, the resulting recursion is a contractive linear part plus residual maps whose total Lipschitz budget is $O(\ell)$. This construction works for all $0 < t \le 1$ when the stated raw-action bound holds.

The normalization multiplies the final readout by at most $\exp(O(\ell))$, so inverse-polynomial accuracy in normalized coordinates still preserves the original sign. These paths admit parallel, rather than layer-by-layer, evaluation. Unrolling the normalized linear part gives a strictly causal equation with propagators T_{ji} of norm at most one. Simultaneous Picard updates take form

$$u_j^{(k+1)} = b_j(x) + \sum_{i<j} T_{ji} F_i(u_i^{(k)}).$$

With $\Lambda = \sum_i \mathrm{Lip}(F_i) = O(\ell)$, the truncation and rounding error is bounded by $C\Lambda^k/k! + \delta e^{\Lambda}$. Consequently $k = O(\ell)$ updates and inverse-polynomial rounding precision preserve the output sign. Scalar lookup, affine arithmetic, radial clipping, and their finite-grid implementations give constant threshold depth per update and polynomial total size. The lookup tables range over polynomial-size scalar grids, not the Boolean input truth table.

Scales bounded away from the critical endpoint. When $a \ge 1/4$, the excess action in (A.1), divided by a, is $O(\ell)$ after depth padding. The entropy two-ball estimate then gives a polynomial-size cover of a normalized prefix after $O(\ell)$ stages. Normalized layer maps are nonexpansive and admit polynomial-size, constant-depth arithmetic approximations. Oppositely labeled representatives are separated by an inverse-polynomial distance, because the remaining network and readout preserve the final margin. An actual threshold circuit computes the prefix and classifies it by squared distances to these representatives. Its size is polynomial and its depth is $O(\ell)$.

Long paths near the critical endpoint. Suppose $0 < a < 1/4$. If $aH \le Q\ell$, the bound $c_t \le a^2/2$ turns (A.1) into the small-raw-action case, with a constant depending only on B' and Q. Otherwise, for a sufficiently large universal Q, scalar last-exit and occupation estimates select a prefix of length $O(\ell/a)$. Its adjusted entropy cost is $O(a\ell)$, and the accumulated variance and square-root entropy budgets are $O(\ell)$. A block-cover argument supplies polynomially many representatives drawn from actual inputs; representatives with opposite labels are separated at inverse-polynomial scale. Although the prefix may have far more than $O(\ell)$ layers, its normalized residual Lipschitz budget is $O(\ell)$, so the same parallel Picard construction evaluates it in $O(\ell)$ threshold depth. The distance classifier completes the circuit. Normalizing the actual Gaussian row gains also keeps the quadrature and rounding resources polynomial independently of t.

Recombination. These cases cover every $0 < t \le 1$, including both boundary values $a = 0$ and $a = 1/4$. Their constants are chosen before the dimension, depth, scale, or sample. Enlarging the common size and depth constants therefore puts every path satisfying (A.1) in one class $\mathcal{H}_R$. The outside-class path extracted above is impossible. Hence the posterior (3) gives $\mathcal{H}_R$ mass at least $5/6$, as required by the main theorem's finite majority reduction.

References

[1] ChatGPT-6 Astra, Kaarel Hänni, and Dmitry Vaintrob. *Frobenius product, parallel circuits, and sample-complexity barriers for deep Gaussian Bayesian networks*. Research manuscript, 2026. Manuscript, source, and proof notes.